Lecture 3 - 01/11

Thursday, November 12, 2026 7:26 PM

Geometry1

Geometry 1:

Pinhole camera, Perspective projection, Projective geometry

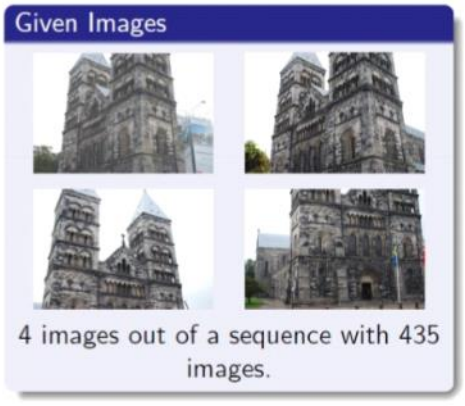
Introduction to Computer Vision

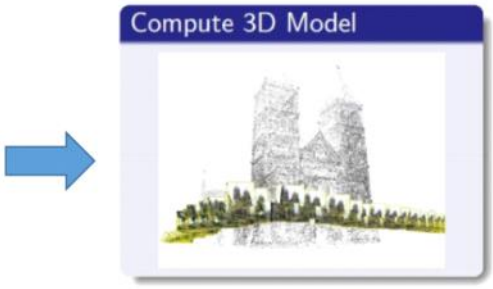
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The aim of geometry in computer vision

- Multi-view reconstruction
- Structure from Motion (SfM)





- Reconstruction of the 3D structure
- Camera pose recovery

Structure from Motion (SfM) Challenges and applications

- The input to the problem is only the 2D images
- The 2D images are (perspective) projections of the 3D scene
- We need to move back from the 2D world to the 3D world
- without any prior knowledge on the 3D scene
- without knowledge on the locations of the cameras

Applications

- Navigation
- Autonomous driving
- Medical imaging, surgery guiding
- Augmented reality
- Photo organization

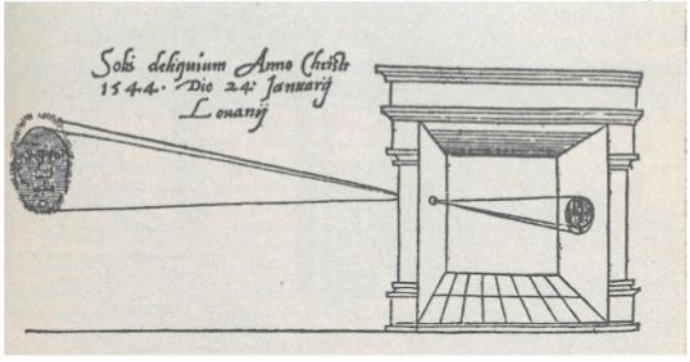
Geometry- we are going to cover

- Pinhole camera model, perspective projection, projective geometry
- Two view geometry, general case:
  - Epipolar geometry, the essential matrix
  - Camera calibration, the fundamental matrix
- Stereo vision
  - 3D reconstruction from two views
- Multi-view geometry
  - reconstruction through factorization

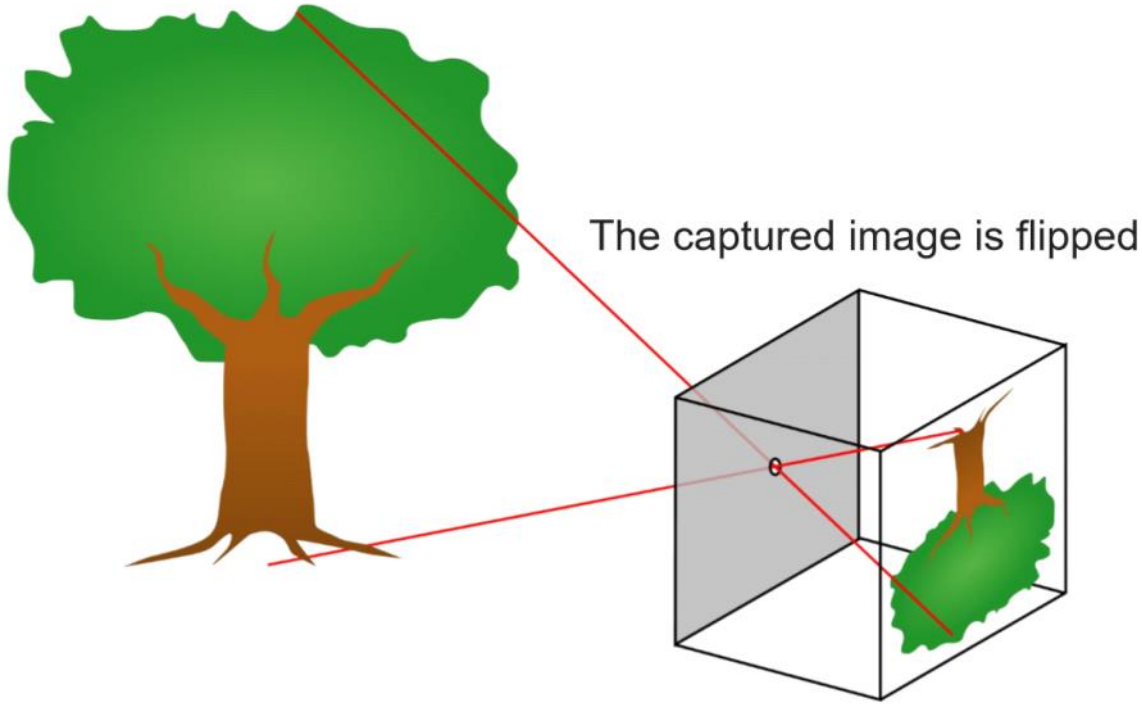
Camera Obscura ("dark room")

- The origin of cameras started a long ago with camera obscura
- A camera obscura is a darkened room with a small hole at one side through which the image is projected onto a wall
- The principle of the camera obscura is called a pinhole camera model
- A simple camera, i.e., a light-proof box with a tiny aperture (pinhole)
- The size of the images depends on the distance between the object and the pinhole

First published picture of camera obscura.
Book *De Radio Astronomica et Geometrica by* Gemma Frisius, 1545 .

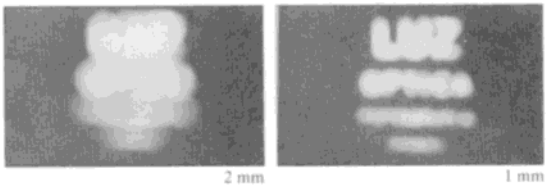


Pinhole camera



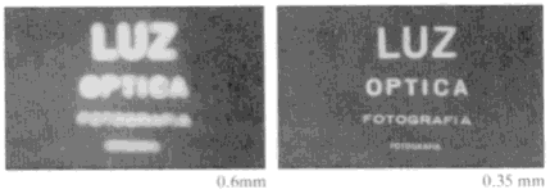
Pinhole camera

- The ideal pinhole camera would have an infinitely small hole, yielding dark image

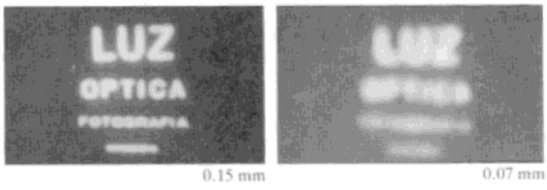


- Pinhole too big – blurry image

- Pinhole is smaller- sharper but dimmer

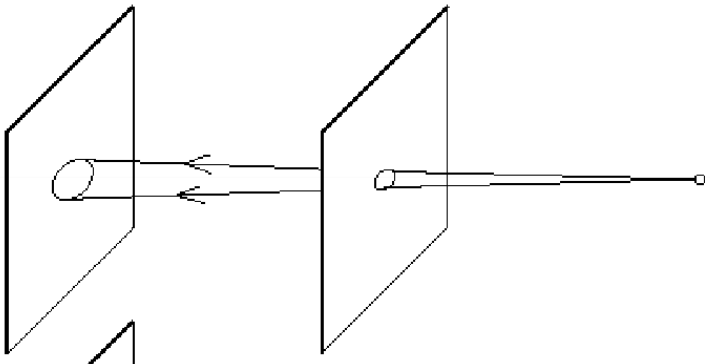


- Pinhole too small – sharpness is lost due to diffraction

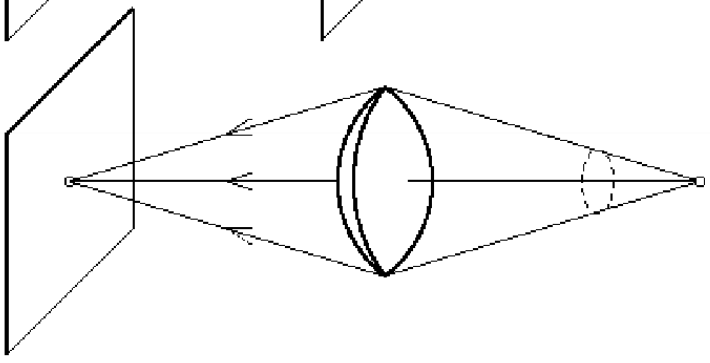


Lenses

Pinhole



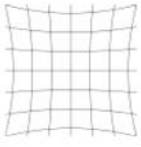
Lens

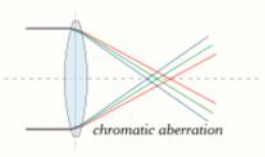


Lenses allow a large aperture yielding a reasonable brightness while maintaining focus

Lenses

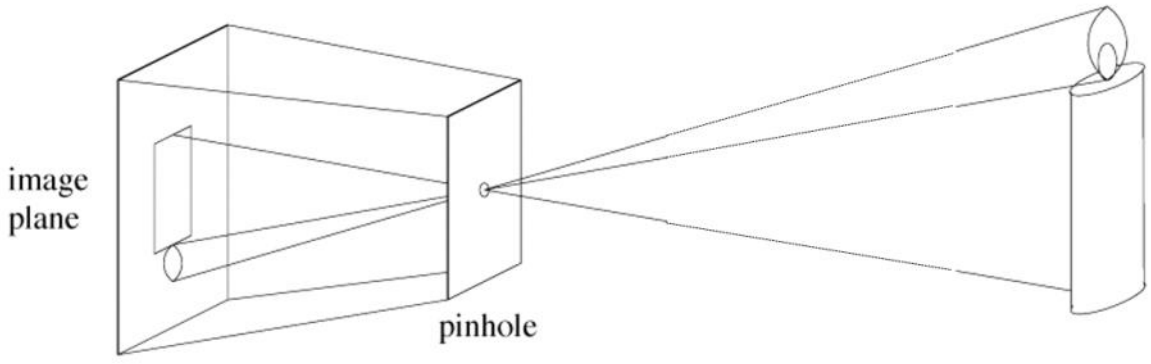
- Lenses collect light from a large hole and direct it to a single point
- Overcome the darkness of pinhole cameras
- However, suffer from
  - Focus tuning
  - Radial distortions
  - Chromatic aberrations





Pinhole camera model

- Pinhole is useful as a mathematical model
- The mapping from three dimensions onto two dimensions is called *perspective projection*
- Perspective: “perspicere” (latine) – to see through

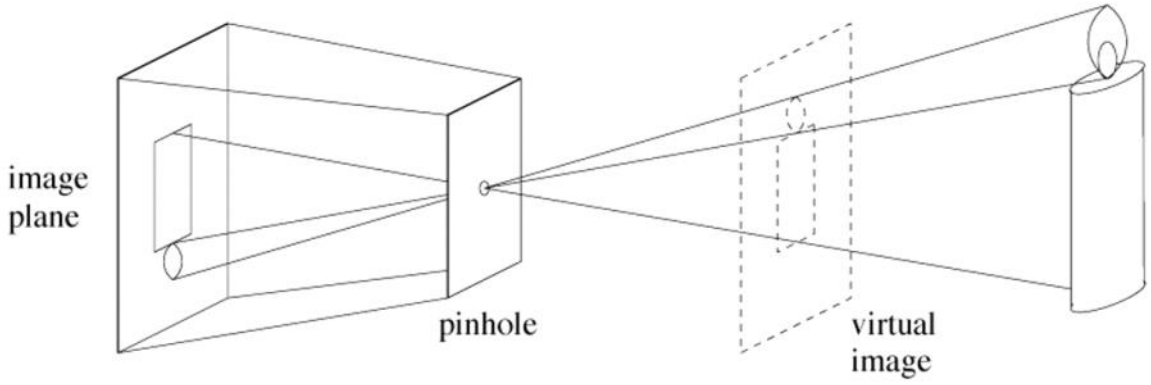


The captured image is flipped.

How to overcome it?

Pinhole camera model

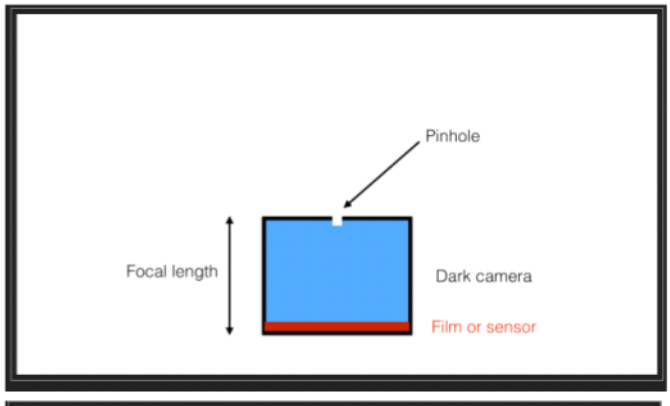
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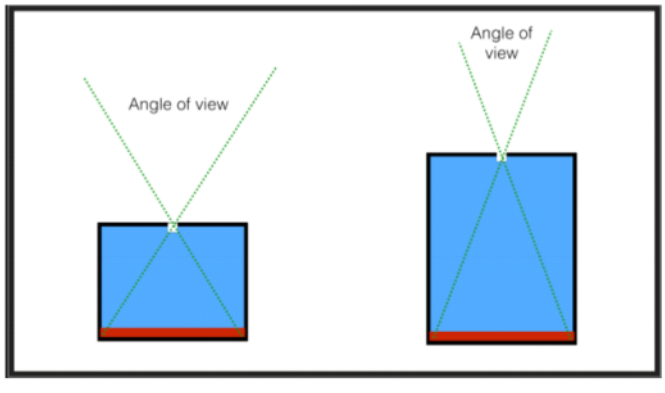


A virtual image plane in front of the pinhole overcomes the image flip

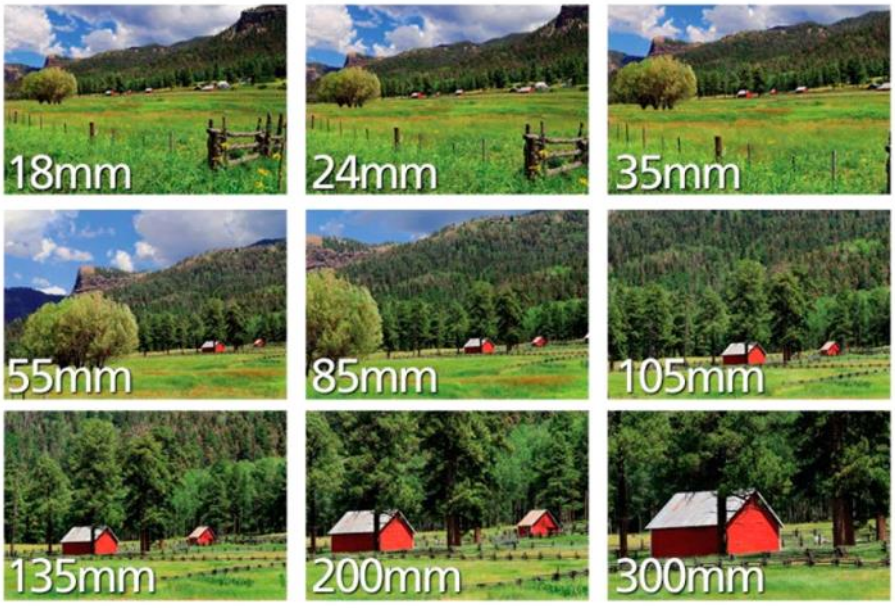
Pinhole camera - focal length

- The depth of the pinhole camera is called the *focal length*.
- The longer the focal length, the narrower the angle of view and the higher the magnification.
- The shorter the focal length, the wider the angle of view and the lower the magnification.





Pinhole camera - focal length

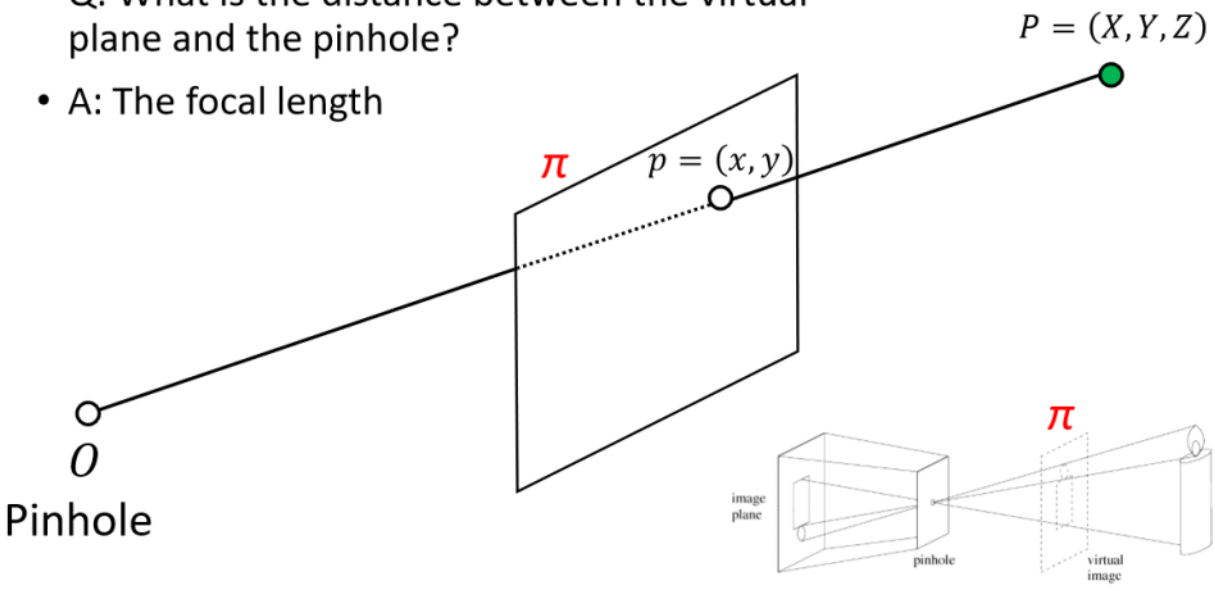


From short focal length to long focal length

Picture by Dave Black

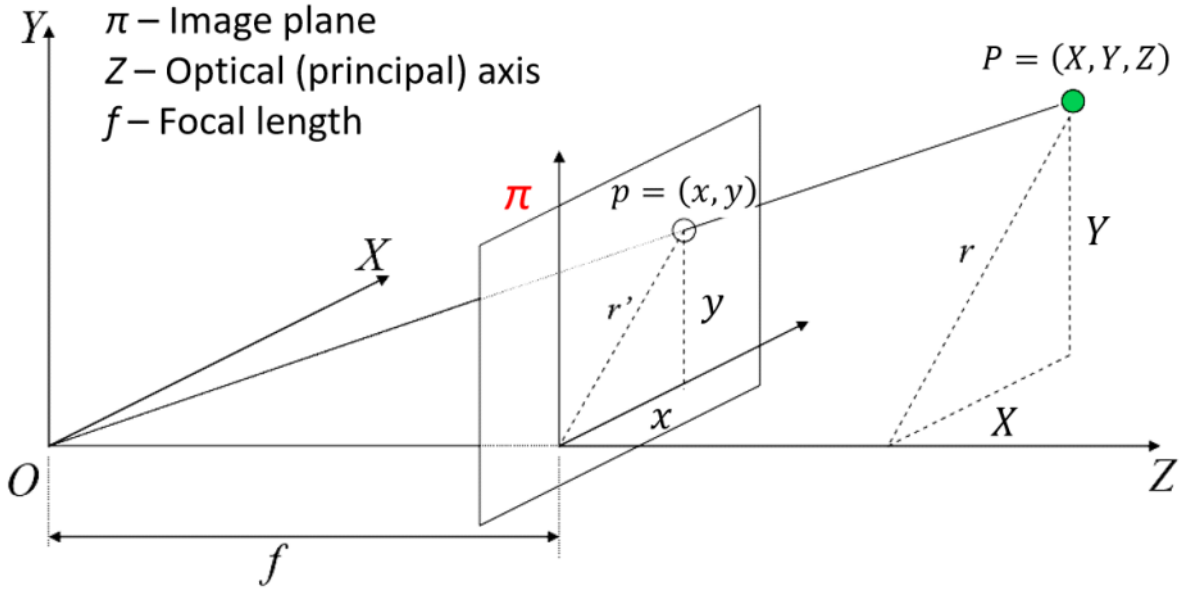
The mathematical model of the pinhole camera
Perspective projection, from 3D to 2D

- A virtual image plane in front of the pinhole
- Q: What is the distance between the virtual plane and the pinhole?
- A: The focal length



The mathematical model of the pinhole camera
Perspective projection, from 3D to 2D

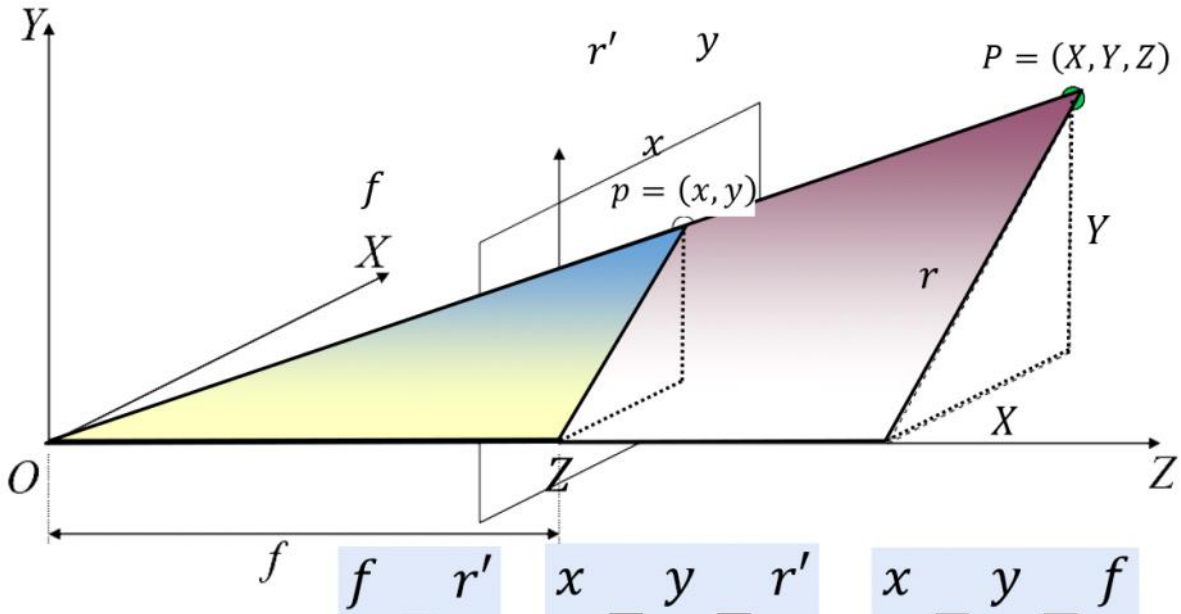
O – Camera (focal) center, $O = (0,0,0)$
π – Image plane
Z – Optical (principal) axis
f – Focal length



(0,0,f) in 3D
|
v
(0,0) in 2D

The mathematical model of the pinhole camera

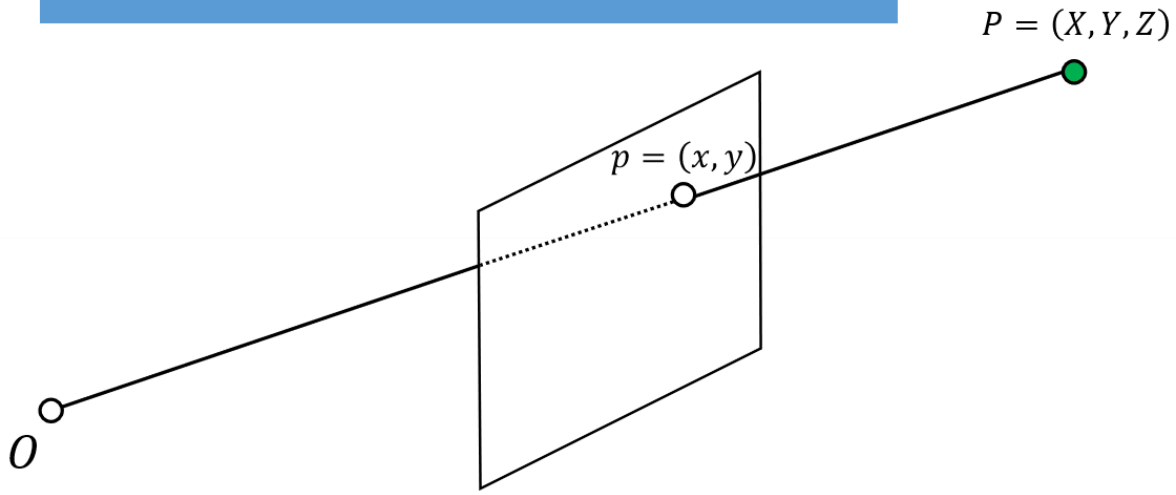
Perspective projection, from 3D to 2D



$$\frac{f}{Z} = \frac{r'}{r} \qquad \frac{x}{X} = \frac{y}{Y} = \frac{r'}{r} \qquad \frac{x}{X} = \frac{y}{Y} = \frac{f}{Z}$$

Perspective rule

$$x = \frac{fX}{Z} \qquad y = \frac{fY}{Z} \qquad z = f$$



- Any 3D point on the line has the same projection
- No unique inverse
- We cannot recover the lost dimension (depth) from a single image

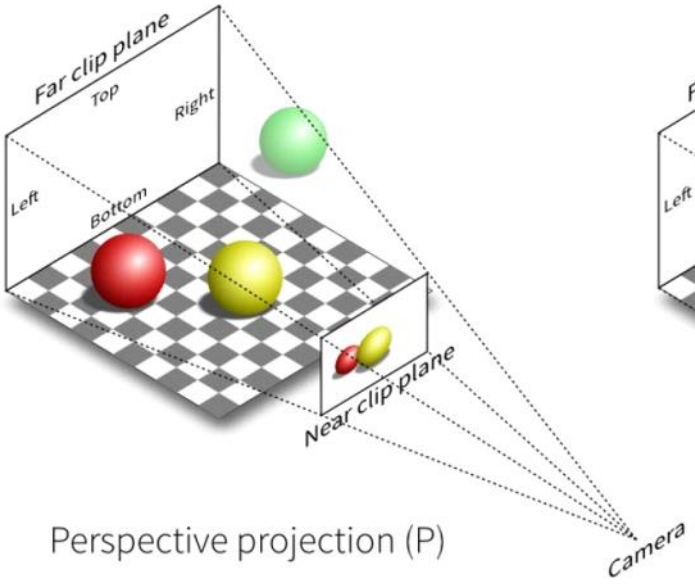
Perspective projection



Lines, distances, angles

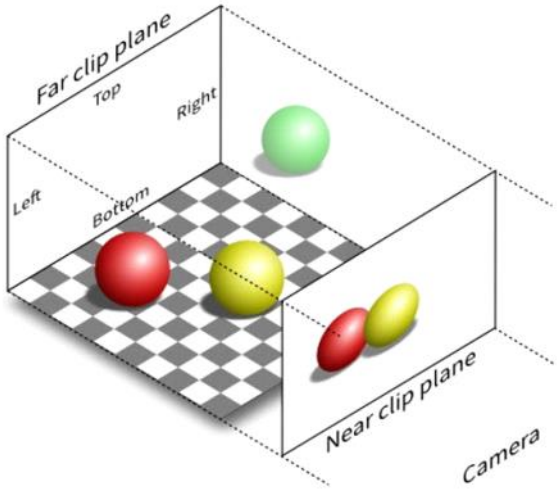
- Lines in 3D project to lines in 2D
- Distances and angles are not preserved
- Parallel lines do not in general project to parallel lines

distant objects appear smaller



Perspective projection (P)

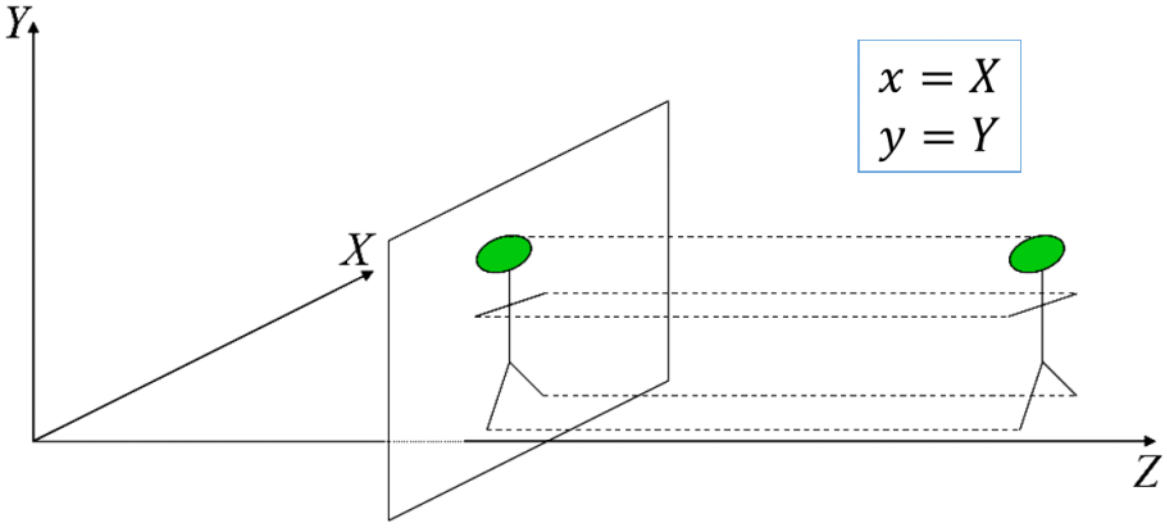
distant objects have the same size as closer ones



Orthographic projection (O)

Orthographic projection

- Projection rays are (nearly) parallel
- Camera center at infinity
- When objects are far from the camera

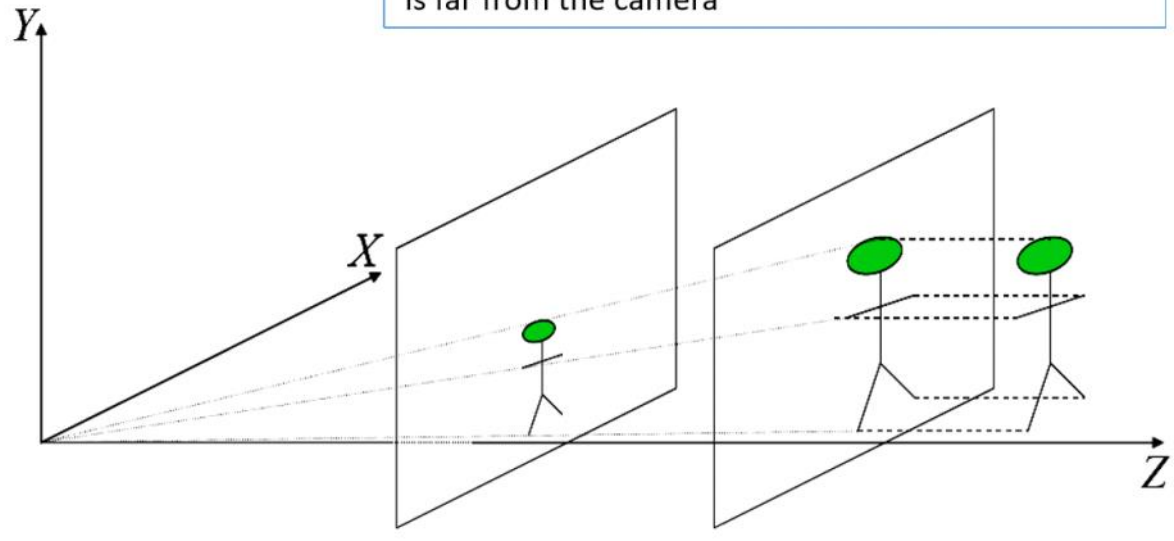


$$x = X$$
$$y = Y$$

Scaled orthographic

$$x = sX \qquad y = sY \qquad s = \frac{f}{Z_0}$$

- Z_0 is some typical distance of the scene
- Useful when the variance of the Z values is small relative to the average depth , e.g., when the object is far from the camera

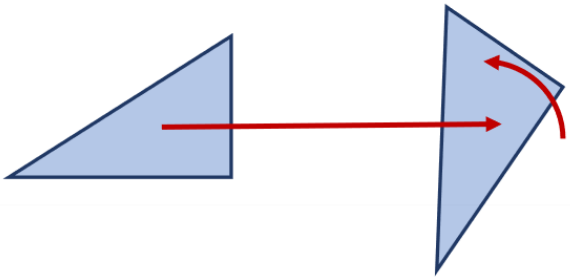


Which projection model should I use?

- Perspective model is needed
  - In scenes that contain many depth differences
  - For accurate 3D reconstruction (stereo, structure from motion)
- Scaled orthographic can be used
  - When objects are small relative to their distance from the camera
  - Often sufficient for recognition applications

Geometry

- Geometry – Greek: earth measurement
- Geometry concerns with shape, size, relative positions, and properties of spaces
- Euclidean geometry:
  - Point, line, plane
  - Incidence – relation between lines and points
  - Preserved under rigid transformations
    - Angles, lengths, areas
    - Parallelism
    - Order, “between”



Euclidean geometry – rigid transformation

- Shapes, Lengths, angles, areas/volumes are preserved under a rigid transformation
- In 2D: (Rotation, 1 dof)

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix}$$

- In 3D: (Rotation, 3 dof)

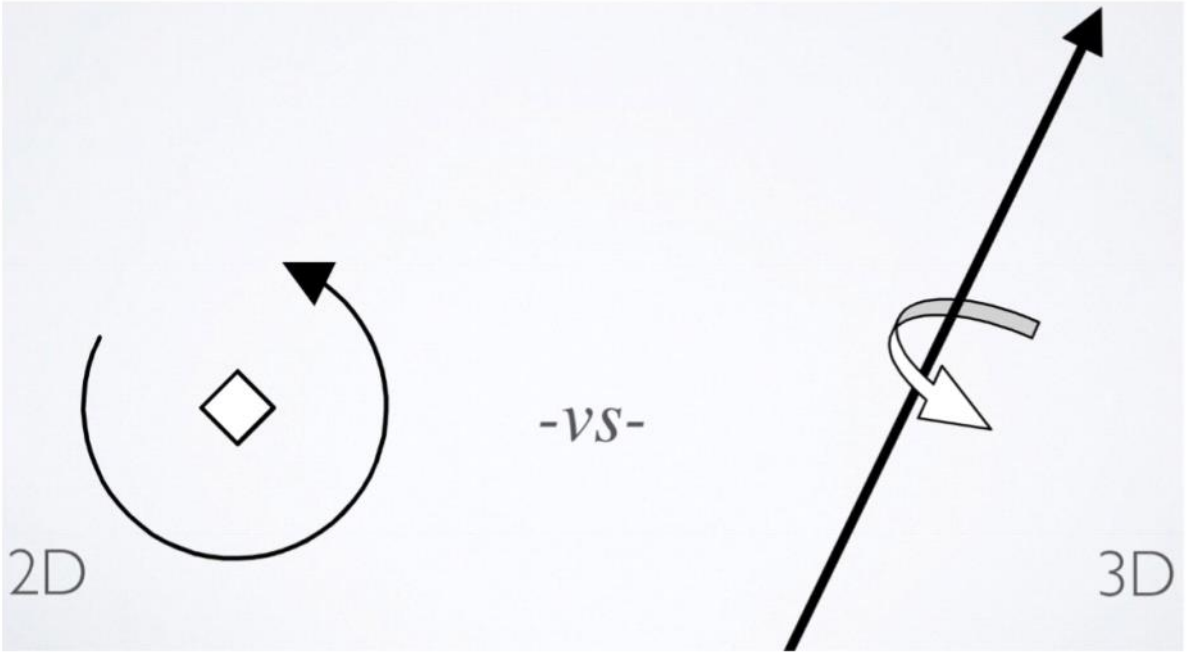
Rotation is represented by a 3×3 matrix R

with $R^T R = I$ and $\det(R) = 1$

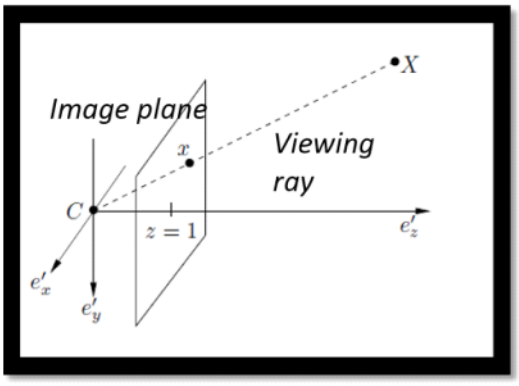
$$\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

3D rotation: axis-angle representation

A unit vector that represents the axis of rotation,
and an angle of rotation about the vector



Projective geometry (motivation)



- Under perspective projection, each point on the viewing ray is projected to the same point x
- In some sense, all the points on the viewing ray are considered as the same point
- *Can we formulate it mathematically?*

Projective geometry (motivation)



- Under perspective projection, parallel lines are not projected as parallel lines
- However, lines are mapped to lines
- *Again, can we formulate it mathematically?*

Projective geometry

- The way to access these observations mathematically is by introducing tools from projective geometry. Specifically,
  - Homogeneous coordinates
  - Points and lines in the projective plane $\mathbb{P}^2$
  - Projective transformations

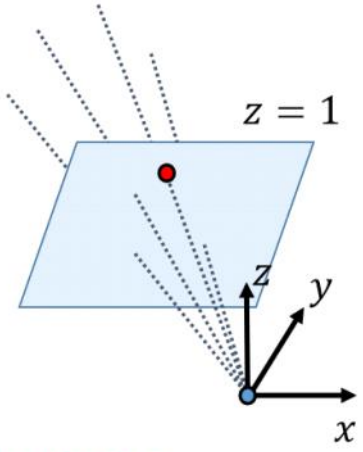
Projective plane $\mathbb{P}^2$

- A **projective plane** is a geometric structure that extends the concept of the Euclidean plane
- In the **Euclidean plane**, two lines are typically intersect a single point, excluding parallel lines that do not intersect
- A projective plane can be considered as an ordinary plane equipped with additional points at infinity, **ideal points**, where parallel lines intersect
  - Given any two distinct points, there is exactly one line incident with both of them
  - Given any two distinct lines, there is exactly one point incident with both of them

Homogeneous coordinates

- In projective geometry we use homogeneous coordinates
- The homogeneous coordinates of a point $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$ are

$$\begin{pmatrix} \alpha x \\ \alpha y \\ \alpha \end{pmatrix} \in \mathbb{P}^2, \alpha \neq 0$$



- Example

$\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 6 \\ 4 \\ 2 \end{pmatrix}, \begin{pmatrix} 9 \\ 6 \\ 3 \end{pmatrix}$, ... are homogeneous coordinates of ...

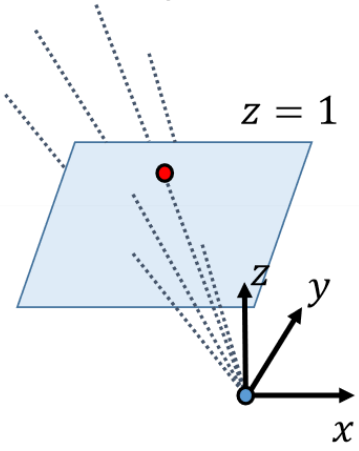
$$\begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 6 \\ 4 \\ 2 \end{pmatrix}, \begin{pmatrix} 9 \\ 6 \\ 3 \end{pmatrix}$, ... are the same element in $\mathbb{P}^2$

- We add a dimension
- All points on a single ray are the same point
- Boil down to a single point in $\mathbb{R}^2$

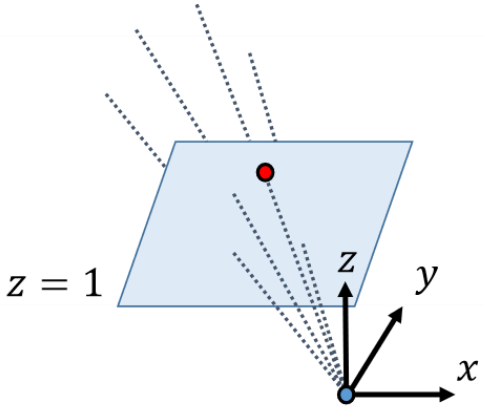
Homogeneous coordinates

- In projective geometry we use homogeneous coordinates
- Adding a dimension, all points on a single ray are the same point, boil down to a single point in $\mathbb{R}^2$
- Enabling non-linear mappings to be represented by linear matrix equations
  - perspective projection
  - projective transformation
- Allowing representation of points at infinity



Projective plane $\mathbb{P}^2$

- The projective space $\mathbb{P}^2$ consists of all the elements that can be represented by vectors in $\mathbb{R}^3 - (0,0,0)^T$
- A point is identified with a ray through the origin, so $\boldsymbol{p}$ and $\alpha\boldsymbol{p}$ for $\alpha \neq 0$ represent the same point
- Projective geometry is a geometry of rays



Homogeneous representation of lines

A line and a point in $\mathbb{P}^2$ have the same algebraic representation

- A line in $\mathbb{R}^2$ is represented as

$$ax + by + c = 0$$

where $(a, b, c)^T \neq (0,0,0)^T$

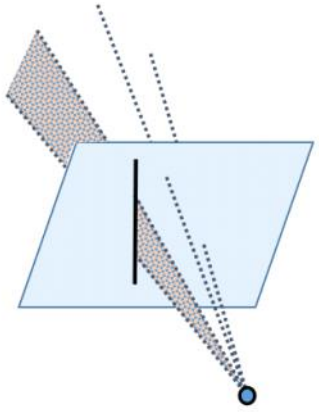
- $\boldsymbol{l} = (a, b, c)^T$ is a representation of the line
- The correspondence between lines and vectors is not one-to-one
- $ax + by + c = 0$ and $(ka)x + (kb)y + (kc) = 0$ are the same lines for any $k \neq 0$
- Thus, $(a, b, c)^T$ and $k(a, b, c)^T$, $k \neq 0$, represents the same line
- $(0,0,0)^T$ does not correspond to any line and therefore is excluded
- A point $\boldsymbol{x} = (x_1, x_2, x_3)^T \in \mathbb{P}^2$ lies on the line $\boldsymbol{l} = (a, b, c)^T \in \mathbb{P}^2$

$$\Leftrightarrow$$

$$\boldsymbol{x}^T\boldsymbol{l} = 0$$

A line in $\mathbb{P}^2$

- A projective line in the projective plane is a set of points
- A point in the projective plane is a ray through the origin
- A line = plane of rays through the origin
- The vector of coefficients l (i.e., "the line l") is normal to the plane of rays



Lines and points in $\mathbb{P}^2$

- The set of equivalence classes of vectors in $\mathbb{R}^3 - (0,0,0)^T$ forms the projective space $\mathbb{P}^2$, where lines and points are represented by homogeneous coordinates
- Points are dual to lines in $\mathbb{P}^2$
- Duality principle. To any theorem of 2 dimensional projective geometry there corresponds a dual theorem, which may be derived by interchanging the roles of points and lines in the original theorem.
- Example. The statement "Two lines intersect each other in one point" is dual to "For any two points there is one line going through them both"

Lines and points in $\mathbb{P}^2$

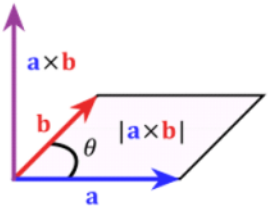
Intersection and incidence

- The intersection of two lines $\boldsymbol{l}_1$ and $\boldsymbol{l}_2$ is the point

$$\boldsymbol{p} = \boldsymbol{l}_1 \times \boldsymbol{l}_2$$

- The line through two points $\boldsymbol{p}_1$ and $\boldsymbol{p}_2$ is

$$\boldsymbol{l} = \boldsymbol{p}_1 \times \boldsymbol{p}_2$$



Proof (of the first claim)

$$(\boldsymbol{l}_1 \times \boldsymbol{l}_2) \perp \boldsymbol{l}_1 \Rightarrow (\boldsymbol{l}_1 \times \boldsymbol{l}_2)^T \boldsymbol{l}_1 = 0 \Rightarrow \boldsymbol{p} \in \boldsymbol{l}_1$$
$$(\boldsymbol{l}_1 \times \boldsymbol{l}_2) \perp \boldsymbol{l}_2 \Rightarrow (\boldsymbol{l}_1 \times \boldsymbol{l}_2)^T \boldsymbol{l}_2 = 0 \Rightarrow \boldsymbol{p} \in \boldsymbol{l}_2$$

$\Rightarrow \boldsymbol{p}$ is the intersection point

Lines and points in $\mathbb{P}^2$

Example. Determine the intersection of the lines $x = 1$ and $y = 1$

First we determine the homogeneous representation of the lines

$$x = 1 \rightarrow -1x + 1 = 0 \rightarrow \boldsymbol{l} = (-1,0,1)^T$$
$$y = 1 \rightarrow -1y + 1 = 0 \rightarrow \boldsymbol{l}' = (0,-1,1)^T$$

$$\Rightarrow \boldsymbol{x} = \boldsymbol{l} \times \boldsymbol{l}' = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{vmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$\Rightarrow$ inhomogeneous point $(1,1)^T$

Ideal points (vanishing points)

- In the space $\mathbb{P}^2$ every pair of lines have a common intersection
- Example. Determine the intersection between the lines $x = 1$ and $x = 2$

$$\boldsymbol{x} = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ 1 & 0 & -1 \\ 1 & 0 & -2 \end{vmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

- This homogeneous point does not have an inhomogeneous representation
- A vanishing point

- Interpretation. The point at infinity in the direction of y-axis
- The point $(0,1,0)^T$ lies on any line of the from $x = a$, represented as $(1,0,-a)$

Ideal points (vanishing points)

- Rays from the origin that are parallel to the $z = 1$ plane define ideal points, $p = (x, y, 0)$. These points do not correspond to an Euclidean point
- How many different ideal points are there?

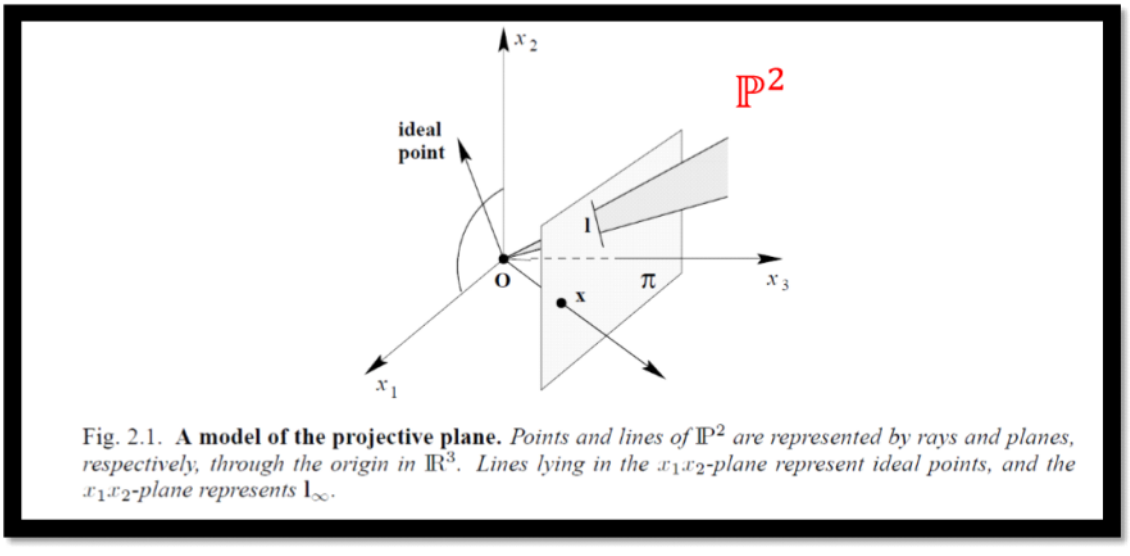
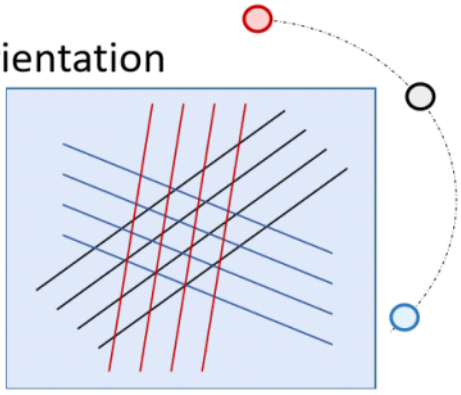


Fig. 2.1. **A model of the projective plane.** *Points and lines of* $\mathbb{P}^2$ *are represented by rays and planes, respectively, through the origin in* $\mathbb{R}^3$. *Lines lying in the* x_1x_2*-plane represent ideal points, and the* x_1x_2*-plane represents* $\mathbf{l}_\infty$.

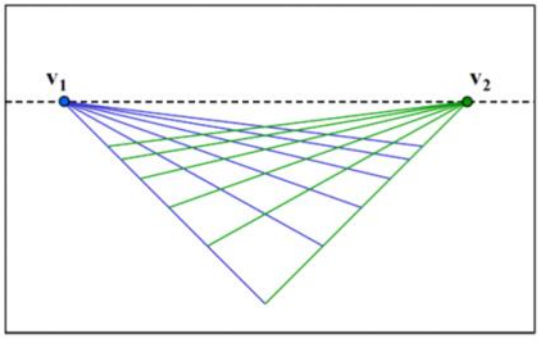
Line at infinity = vanishing line

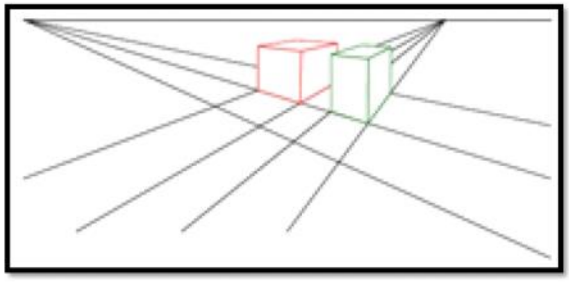
- Parallel lines intersect at an ideal point:
  $l_1 = (a, b, c)^T$ and $l_2 = (a, b, c')^T$ intersect at
  $$p = l_1 \times l_2 \propto (b, -a, 0)$$
- Every set of parallel lines of different orientation intersects at a different point at infinity
- The complete set of ideal points is called the line at infinity
- The line at infinity is indeed a line = a plane of rays through the origin (the $Z = 0$ plane)
- A line at infinity can be thought of the set of "directions" of lines in the plane
- Algebraically, $l_\infty = (0,0,1)^T$



Vanishing lines and vanishing points

- Any set of parallel lines on the plane define **a vanishing point**
- The union of all vanishing points from lines **on the same plane** is the vanishing line
- For the **ground plane**, this is called the **horizon**





Projective transformation = Projectivity = Homography

A projectivity is an invertible mapping from points in $\mathbb{P}^2$ (homogeneous 3-vectors) to points in $\mathbb{P}^2$ that maps lines to lines.

Def. A projective transformation is an invertible mapping h: $\mathbb{P}^2 \rightarrow \mathbb{P}^2$ such that three points $\boldsymbol{x_1}, \boldsymbol{x_2}$ and $\boldsymbol{x_3}$ lie on the same line if and only if $h(\boldsymbol{x_1}), h(\boldsymbol{x_2})$ and $h(\boldsymbol{x_3})$ are on the same line

Projective transformation = Projectivity = Homography

Representation of homography

Theorem. A mapping $h: \mathbb{P}^2 \to \mathbb{P}^2$ is a projective transformation if and only if there exists a non-singular matrix 3×3 matrix H such that for any point $\boldsymbol{x} \in \mathbb{P}^2$ it holds that $h(\boldsymbol{x}) = H\boldsymbol{x}$

Proof. (partial)

Let $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3$ lie on a line $\boldsymbol{l}$, i.e., $\boldsymbol{l}^T\boldsymbol{x}_i = 0$, for $i = 1,2,3$. Let H be a non-singular 3×3 matrix. Since $\boldsymbol{l}^T H^{-1} H\boldsymbol{x}_i = 0 \Rightarrow$ the points $H\boldsymbol{x}_i$ lie on the line $H^{-T}\boldsymbol{l}$.

Homography transforms a line to a line

Projective transformation = Projectivity = Homography

- An invertible linear transformation of rays

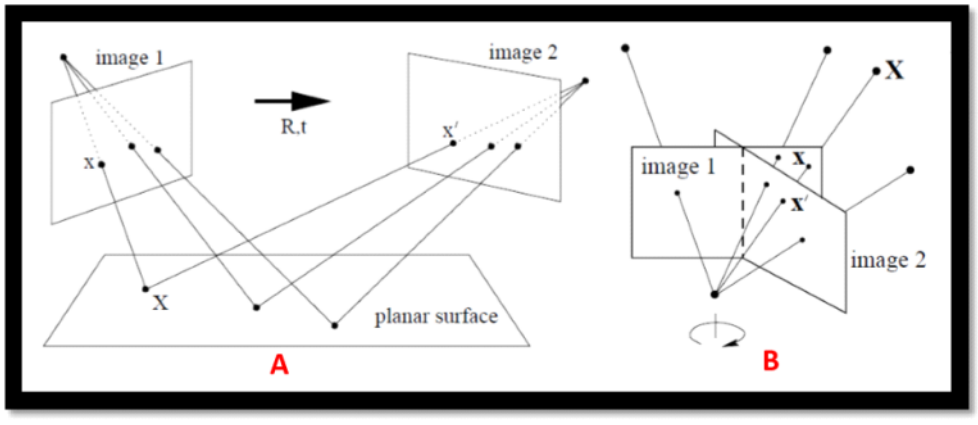
$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \boldsymbol{q} \propto H\boldsymbol{p} = \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

- Non-linear transformation of points

$$x' = \frac{h_{11}x+h_{12}y+h_{13}}{h_{31}x+h_{32}y+h_{33}}, \quad y' = \frac{h_{21}x+h_{22}y+h_{23}}{h_{31}x+h_{32}y+h_{33}}$$

- Homography has 9 entries, only 8 DOFs, why?
- H is determined up to scale
- Homography can be calculated by Direct Linear Transform
- How many points are needed to determine homography?
  - 4 points: each point determines two equations

Examples of projective transformation

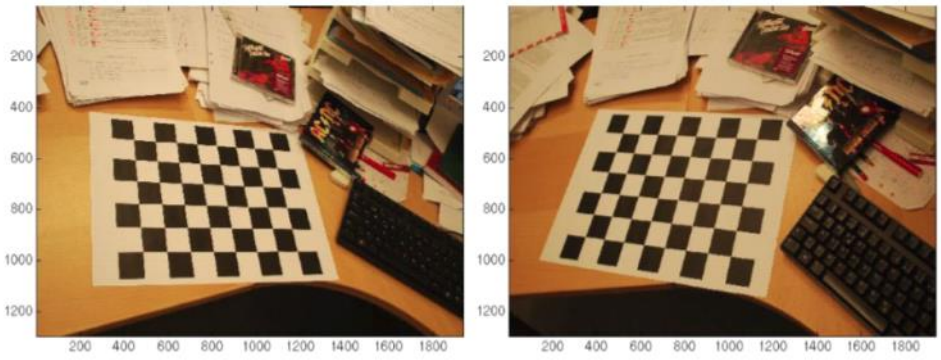


(A) The projective transformation between two images induced by a world plane

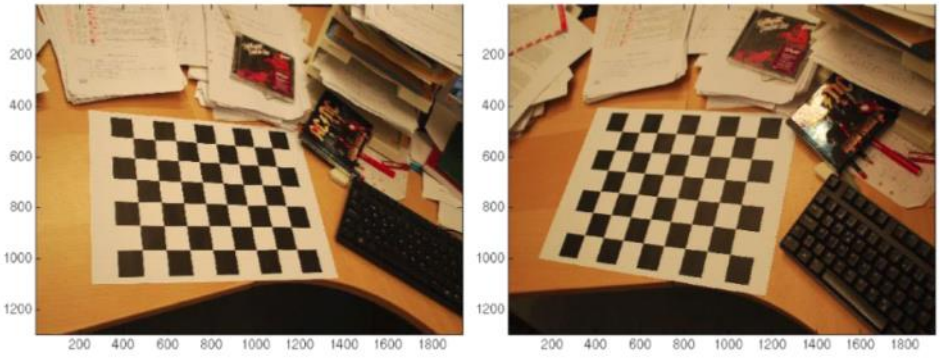
(B) The projective transformation between two images with the same camera center (camera rotating about its center, no translation)

- Proof in a few slides

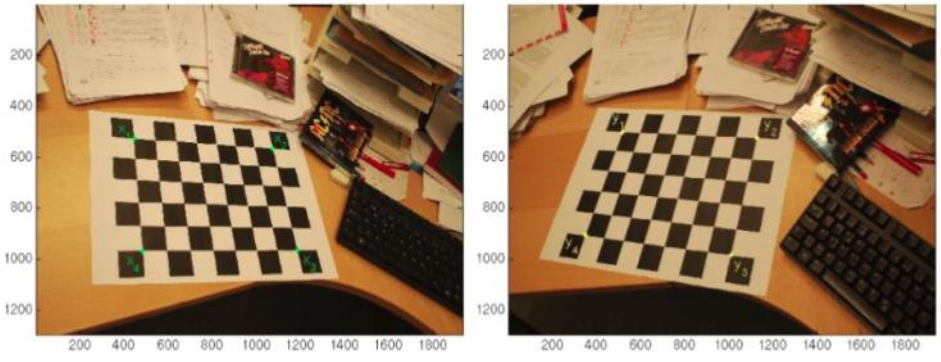
Homography example
(A) Two cameras captured a world plane



- A set of points $\boldsymbol{U}_i$ lying on the same plane is projected into two cameras
- $\boldsymbol{x}_i \sim C_1\boldsymbol{U}_i$ and $\boldsymbol{y}_i \sim C_2\boldsymbol{U}_i$ then there is a homography such that $\boldsymbol{x}_i \propto H\boldsymbol{y}_i$
- Calculate H, how many degrees of freedom?

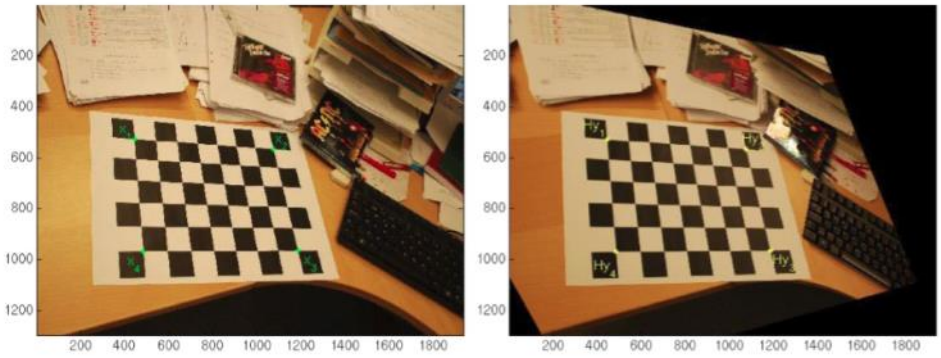


- There are 9 elements in H, however the scale does not matter (H and λH represents the same projective transformation), so we are left with 8 degrees of freedom.
- Now, how may equations do we need to solve H?

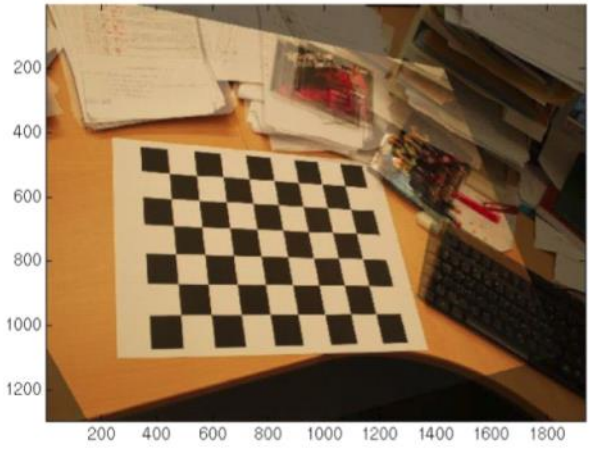


- Suppose, we know that n points $\boldsymbol{y}_i$, $i = 1, \dots, n$ are transformed to $\boldsymbol{x}_i$
- By the relation $\boldsymbol{x}_i \propto H\boldsymbol{y}_i$ each pair contributes 2 equations
- So, in total we have $2n$ equations and 8 degrees of freedom

$$2n \geq 8 \Leftrightarrow n \geq 4$$



Result of applying homography to the right image



Mean value of the two images.

The points on the plane seem to agree.

Projective transformation

- Under projective transformation
  - A (straight) line transforms to a line and a conic to a conic
  - But order and parallelism are not preserved
  - Likewise, angles, lengths and areas are not preserved

In contrast to rigid transformation in Euclidean geometry

Homography: main points

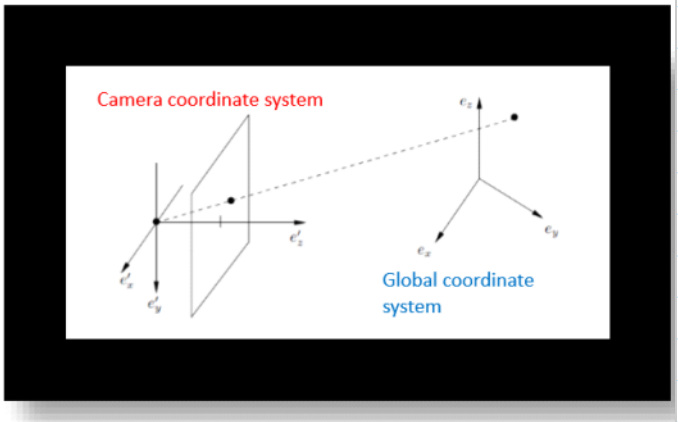
$$\lambda \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

(A) If the scene is planar, then the images from any two cameras are related by a homography

(B) If camera rotates about its center, then the images are related by a homography irrespective of scene depth

Homography mapping is a 3×3 invertible matrix with 8 dof

Camera motion model



$$\begin{bmatrix} X_1' \\ X_2' \\ X_3' \end{bmatrix} = R \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} + \boldsymbol{t}$$

$R \in \mathbb{R}^{3\times 3}$ rotation matrix
$R^T R = I, \ \det(R) = 1$
$\Rightarrow \mathrm{R}^{-1} = R^T$
$\boldsymbol{t} \in \mathbb{R}^3$ translation vector

$$X_3' \begin{bmatrix} X_1'/X_3' \\ X_2'/X_3' \\ 1 \end{bmatrix} = \begin{bmatrix} X_1' \\ X_2' \\ X_3' \end{bmatrix} = [R \ \boldsymbol{t}] \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ 1 \end{bmatrix}$$

*homogeneous coordinates of a 3D point

The rows of R encodes the x' axis, y' axis and z' axis of the local coordinate system, as expressed in the global coordinate system

(B) Camera rotation ➔ homography

$$\boldsymbol{Q} = R\boldsymbol{P} + \cancel{\boldsymbol{t}}$$

$\boldsymbol{p}, \boldsymbol{q}$ are perspective projections of $\boldsymbol{P}, \boldsymbol{Q}$

$$\boldsymbol{p} = \frac{f}{Z}\boldsymbol{P} \Rightarrow \boldsymbol{P} = \frac{z}{f}\boldsymbol{p}$$

$$\boldsymbol{q} = \frac{f}{Z'}\boldsymbol{Q} \Rightarrow \boldsymbol{Q} = \frac{z'}{f}\boldsymbol{q}$$

Substitute in $\boldsymbol{Q} = R\boldsymbol{P}$

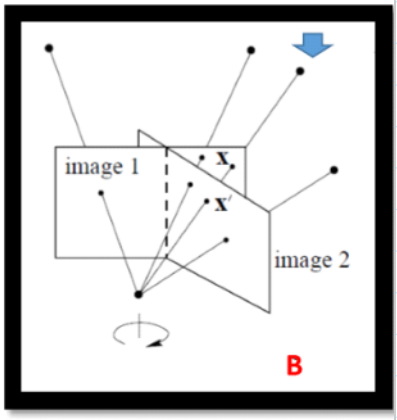
$$\frac{z'}{f}\boldsymbol{q} = \frac{z}{f}R\boldsymbol{p}$$

$$\Rightarrow \boldsymbol{q} = \frac{Z}{Z'}R\boldsymbol{p}$$

$$\Rightarrow \boldsymbol{q} \propto R\boldsymbol{p}$$

(need to verify that the scaling is free of 3D information)

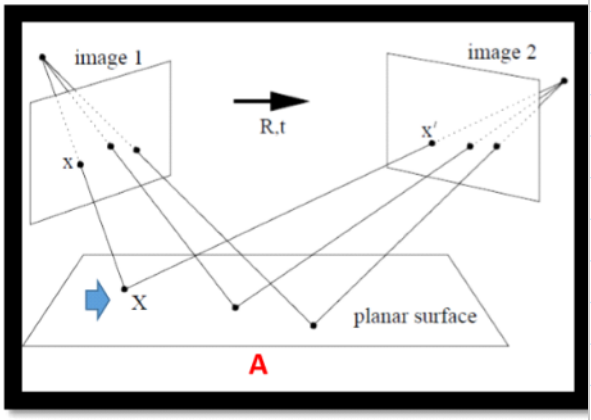
Rotation around the focal center (no translation)



$$\boldsymbol{P} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \quad \boldsymbol{Q} = \begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix}$$

(A) Planar scene ➔homography

- A plane $aX + bY + cZ = d$
- In vector notation $\boldsymbol{n}^T \boldsymbol{P} = d$
- $\boldsymbol{Q} = R\boldsymbol{P} + \boldsymbol{t}$
- $\frac{\boldsymbol{n}^T \boldsymbol{P}}{d} = 1$, so $\boldsymbol{Q} = R\boldsymbol{P} + \boldsymbol{t}\frac{\boldsymbol{n}^T \boldsymbol{P}}{d}$
- Rearrange, $\boldsymbol{Q} = \left(R + \frac{1}{d}\boldsymbol{t}\boldsymbol{n}^T\right)\boldsymbol{P}$
- Project, $\boldsymbol{q} \propto \left(R + \frac{1}{d}\boldsymbol{t}\boldsymbol{n}^T\right)\boldsymbol{p}$
- So $\boldsymbol{q} \propto H\boldsymbol{p}$, with $H = R + \frac{1}{d}\boldsymbol{t}\boldsymbol{n}^T$



$$\boldsymbol{P} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \quad \boldsymbol{Q} = \begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix}$$

Properties of projective transformations

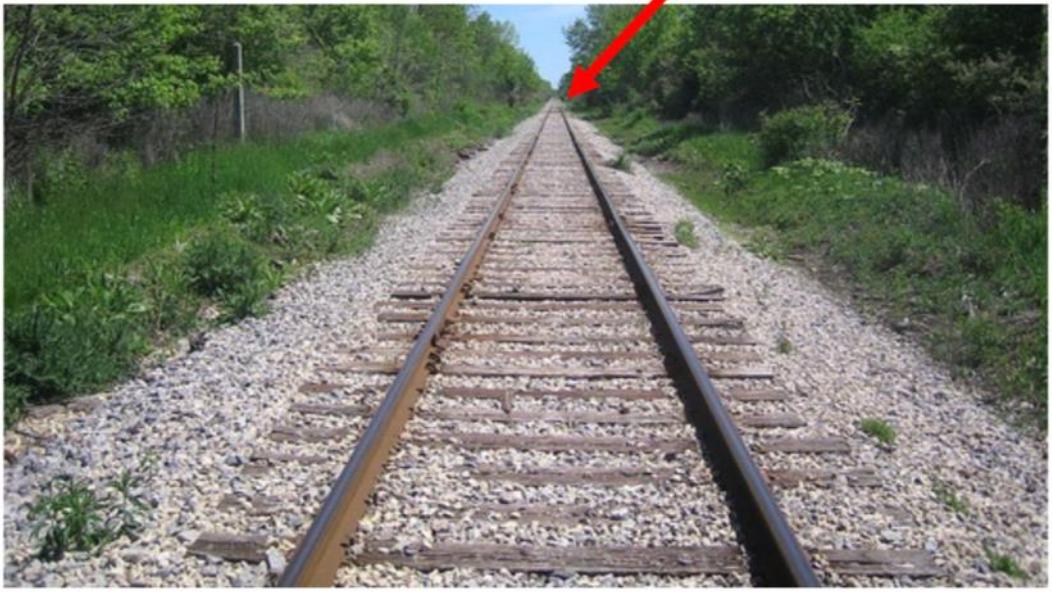
- Points project to points
- Lines project to lines
- Distant objects look smaller



Properties of Projection

- Angles are not preserved
- Parallel lines meet!

Parallel lines in the world intersect in the image at a "vanishing point"



Horizon line (vanishing line)



Hierarchy of transformations

Rigid	$\begin{pmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{pmatrix}$	Preserves angles, lengths, area, parallelism
Similarity	$\begin{pmatrix} a & -b & t_x \\ b & a & t_y \\ 0 & 0 & 1 \end{pmatrix}$	Preserves angles, parallelism
Affine	$\begin{pmatrix} a & b & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{pmatrix}$	Preserves parallelism
Homography	$\begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix}$	(up to scale) Preserves cross ratio